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Behavioural Economics Series 2: Are Financial Markets Efficient?

The 2013 Nobel Prize in Economics was definitely a “weird” one. One recipient was Dr. Eugene Fama from UChicago, who proposed the Efficient Markets Hypothesis (EMH) (which contends that markets are efficient). Yet, on the other hand, professor Robert Shiller from Yale, a leading economist in Behavioural Finance (an area that explores how

psychology engenders investors' decisions to deviate from perfect rationality), also received the honor for his work in examining the irrationality of investors, among some of his other work [1].

It is definitely interesting to see that two academics whose works have generated opposite views on market efficiency were awarded the Nobel Prize in the same year. Both Fama and Shiller are prominent scholars who fully deserve the award, yet their theories generate very different implications. This article aims to briefly explore the justifications of both Fama's and Shiller's viewpoints.

The Efficient Market Hypothesis

What exactly does an "efficient market" mean? In plain words, EMH contends that since there are so many participants, financial markets fully reflect all available information to investors.

There are, in fact, three forms of market efficiency. Weak-form EMH argues that the current market price reflects all past price movements, so what technical analysts do – analyzing graphs and charts of past price and volume data – does not improve stock picks at all. The semi-strong form of EMH states that the market price also reflects all publicly available information, including company's potential profitability and risk level, which means that fundamental analysis – using various financial analysis techniques to value a company's inherent value – is also useless. Furthermore, strong-form EMH, the most stringent form, argues that even insider information is also included in the market [2].

As a very controversial theory, EMH has a lot of evidence that supports its validity as well as numerous counterpoints against it. On one hand, we can often observe how speedy markets process information and reflect it in the prices of the securities. For instance, when Trump recently announced an import tariff on aluminum and steel, the Dow

Jones Industrial Average dropped by 400-500 points almost instantaneously because of the changes in the investors' expectation on the economy. The stock market speedily processes the concerns for a trade war and numerous political turmoils that would engender less economic growth, which in turn leads to lower profits for the companies. The mere fact that markets arrive at a newly corrected price that reflects these concerns in just a few minutes or hours illustrates the speed (and thus efficiency) of how markets respond and process new information. Such efficient responses also evident in the financial markets' to most newly released public information, both economy-wide and company specific.

On the other hand, is that newly arrived price the "right" price? In many cases, it may be observed that markets are also very inefficient in terms of reflecting a "correct" or "rational" price. For one thing, it has almost been a consensus that strong-form market efficiency doesn't exist, since otherwise insider trading (a not only unethical but also illegal behavior), which has been notoriously lucrative, wouldn't even profit. Moreover, there have been an enormous amount of mispricings of financial securities, some of which (small-cap stocks) were even admitted by Fama himself (although that doesn't change his support of the EMH). One simple example of asset mispricing would be the recent hype of cryptocurrencies, where the price doesn't really reflect the intrinsic value of the listed coins.

Shiller and Behavioural Finance

The EMH is definitely a theoretical breakthrough that establishes the definition of an efficient market, but ultimately, it is just a hypothesis and a way too idealistic simplification of the reality. Therefore, in the recent years, experts in Behavioural Finance (a field that was literally heresy at EMH's prime), like Robert Shiller, have received growing attention from both the academia and the public.

Professor Robert Shiller's work has mainly concentrated in exploring where markets are not efficient and explore the irrationality of the market participants by incorporating concepts from sociology and psychology. On one hand, Shiller argued that market prices often don't reflect the expectations that should have been formed under an efficient market. Shiller is an expert in studying the formation of asset bubbles and has successfully predicted the burst of the Dot Com Bubble as well as the housing bubble burst that caused the Great Recession of 2008.

In the late 1990s, a Dom Com Bubble took place and the stock prices of tech companies (mostly internet companies listed on Nasdaq) increased dramatically (which is analogous as how cryptocurrency has emerged in the past two years). Shiller and Greenspan, the Fed Chair at the time, characterized this phenomenon as an "irrational exuberance" that was not going to sustain long since the intrinsic values of these companies' stocks were not nearly close to the height at which their stocks had been traded to. Shiller went further and wrote a book named *Irrational Exuberance* to analyze the causes of the sharp asset price increase and better policies to manage the potential bubble. Coincidentally, the book was published in March 2000, just before the bubble actually burst and Nasdaq crashed. History does repeat itself. In the second edition of *Irrational Exuberance* in 2005, Shiller, again, successfully predicted how the housing bubble. Housing prices, just like dot-com stocks, formed a huge asset bubble as evident from the Case-Shiller index (a housing price index), which eventually burst and engendered the subprime crisis.

The theoretical justification that Shiller employs to explain why market participants are irrational and assets are often mispriced is the concept of mass psychology, as he claims "mass psychology is likely the prominent cause of movements in the price of the aggregate stock market" [3]. In fact, economists as early as Keynes has explored how irrational psychological factors, which Keynes characterized as "animal spirits", impact investors' decision, lead to market panicking and cause

financial crises. The logic behind Shiller's view, in simpler terms, argues that if a lot of investors in the markets are anticipating a drop soon, the market will actually "crash." Such concerns of bubble bursts tend to increase dramatically before bubbles actually burst, which is evident from headings of newspaper articles. The same logic applies when people are "hyping" assets prices on the way up, without any regard to the actual, or intrinsic value of the assets. In one of Shiller's recent articles on the NYTimes, Shiller again expresses his view that the current market is vulnerable (with assets such as cryptocurrencies), as newspaper articles are demonstrating increasing concerns of the burst of a crypto bubble [4].

In my view, EMH does build a theoretical framework under which an ideal market should operate in, and there is a lot of truth to it in terms of the speed of processing new information. Yet it is far from how markets work in reality. As has been proven empirically by economists such as Shiller, markets are not efficient in many regards, securities are often mispriced and do not fully reflect all available information, which is also obvious from previous asset bubble bursts. EMH is a complete framework of an idealistic market, whereas Behavioural Finance is a rather undeveloped field that focuses on how people actually make decisions (in particular, irrational decisions) using theories from psychology and sociology. Hopefully one day, there will be a theory that is a synthesis of these views to paint a complete and accurate picture of how financial markets behave.

References:

[1] <https://www.newyorker.com/news/john-cassidy/inefficient-markets-a-nobel-for-shiller-and-fama>

[2] Bodie, Zvi, et al. Investment. 10th ed., Irwin, 2014.

[3] Shiller, R. J. (2003) 'From Efficient Markets Theory to Behavioral Finance', Journal of Economic Perspectives 17(1): 83–104.

[4] <https://www.nytimes.com/2017/09/15/business/stock-market-mass-psychology.html>

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